



Course Syllabus – Spring Semester 2025
Software Engineer Career Starter

1. Course Information

Catalog Description	This course provides a practical, hands-on approach to modern DevOps practices used in enterprise environments. It focuses on the key tools, workflows, and methodologies required to build, deploy, and operate scalable software systems. The course introduces core DevOps concepts and tools including version control, infrastructure as code, configuration management, containerization, and orchestration. It covers monitoring practices, as well as CI/CD pipelines for automated build, testing, and deployment across multiple environments. The course also explores topics such as cloud infrastructure provisioning, system reliability, troubleshooting, and DevOps automation techniques.
Credit Hours	3
Prerequisite	
Course Type	Lecture
Required/Elective	Elective
Textbook	
References	
Instructor Name	Ayat Barakat AlBqour
Instructor Email	Ayat.albqoor@digitinary.com
Class Schedule	Mon Wed 02:00 - 03:30
Class Location	
Office Hours	Mon Wed Mon Wed
Teaching Assistant	-

2. Course Contents

Week	Topic(s)	Chapter in Textbook
1+2	Introduction to SDLC and DevOps Work with Linux OS Manage source code using Git	
2+3+4+5	Introduction to Docker Practical Workshop (1): Deploy 3-tier Application using Docker	
6+7	Introduce a CICD pipeline Practical Workshop (2): Design and implement simple CICD pipeline	
8	Midterm Exam	
9+10+11	Introduction to Kubernetes and Helm Practical Workshop (3): Deploy Django application using k8s and Helm	
12+13+14	Practical Workshop (3): Integrate Docker , k8s , Helm and CICD pipeline	
15	Project presentation and evaluation	
16	Final Exam	

3. Course Learning Outcomes:

Upon completion of this course, students will be able to:

1. Understand the foundational concepts of the Software Development Life Cycle (SDLC) and DevOps.
2. Work efficiently with Linux-based operating systems
3. Learn Basic Posix systems structure and commands.
4. Learn how to manage the source code using Git
 - a. Understand the concepts of versioning
 - b. Create a Git repository
 - c. Secure repository access
 - d. Learn how to use git commands: add, commit, checkout, merge, rebase
5. Understand and apply containerization using Docker
 - a. Install and configure Docker on Linux
 - b. Build efficient Docker images using Dockerfiles
 - c. Manage containers, volumes, networks, and ports
 - d. Publish and pull images using Docker Hub or private registries
 - e. Use Docker Compose to orchestrate multi-container systems
 - f. Deploy a complete 3-tier application using Docker in a practical workshop
6. Design and implement Continuous Integration and Continuous Delivery (CI/CD) pipelines

- a. Understand CI/CD concepts and role in DevOps automation
- b. Design simple CI pipelines for building and testing applications
- c. Implement CD pipelines for automated application deployment
- d. Use popular CI/CD tools (GitHub Actions / GitLab CI)
- e. Integrate container builds, security checks, and automated tests into the pipeline
- 7. Understand container orchestration concepts using Kubernetes
 - a. Explain Kubernetes architecture (nodes, pods, services, controllers)
 - b. Deploy applications to Kubernetes clusters
 - c. Manage ConfigMaps, Secrets, Services, and Ingress resources
 - d. Understand Stateful vs Stateless deployments
 - e. Understand different deployment strategy
- 8. Package and deploy applications using Helm
 - a. Understand Helm charts, templates, and values files
 - b. Customize deployments through Helm configuration
 - c. Package and publish Helm charts to a repo
 - d. Deploy a Django application using Kubernetes and Helm
- 9. Integrate Docker, Kubernetes, Helm, and CI/CD
 - a. Build a full end-to-end deployment workflow
 - b. Automate image building, pushing, and deployment to Kubernetes
 - c. Implement GitOps-style workflows for consistent deployments
 - d. Develop a full DevOps pipeline for a real application in Practical Workshop (3)

4. Assessment Policy

Assessment Tool	Expected Due Date	Weight
Activities (Quizzes, Homework, Projects, Coursera)	All Course duration	30%
Mid Exam	TBD	30%
Final Exam	TBD	40%

5. Contribution of the Course to the Professional Component

Software Engineering Topics	100%
General Education	0%
Mathematics & Basic Sciences	0%

6. Expected level of proficiency from students entering the course

Mathematics	Not applicable
Physics	Not applicable

Technical writing	Fair
Computer programming	High

7. Material available to students, instructors, TAs, and department at the end of the course

	Students	Department	Instructors	TA(s)
Course objectives and outcomes form	X	X	X	
Lecture notes, homework assignments, and solutions	X	X	X	
Samples of homework solutions from 3 students		X		
Samples of lab reports of 3 students				
Samples of exam solutions from 3 students		X		
Course performance forms from student surveys		X	X	
End-of-course instructor survey		X	X	